ASSIGNMENT 1

COURSE NAME:

ENVIRONMENTAL AND SUSTAINABILITY ANALYTICS

COURSE CODE: ECCSS 20

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ROLL NO.: M2024ECCSS003

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1. INTRODUCTION

Rainfall variability represents one of the most critical determinants of agricultural productivity, water resource availability, and socio-economic stability in India, where approximately 68% of the population depends on agriculture and nearly 55% of cultivated land remains rainfed. The monsoon-driven climate of India exhibits substantial spatial and temporal heterogeneity, with annual rainfall ranging from less than 100 mm in the arid western regions to over 11,000 mm in parts of the northeastern states. This inherent variability, compounded by increasing climate change impacts, makes India particularly vulnerable to meteorological droughts—prolonged periods of below-normal precipitation that can trigger cascading agricultural, hydrological, and socio-economic crises.

Historical evidence demonstrates that India experiences drought conditions in some part of the country nearly every 2-3 years, with severe droughts affecting approximately 16% of the country's area on average. The devastating impacts of major droughts, such as those in 2002-2003, 2012, and 2015-2016, underscore the urgent need for comprehensive assessment of long-term rainfall patterns and drought probability across diverse climatic regions. Understanding the spatial distribution of rainfall variability and drought risk is essential for developing effective drought mitigation strategies, optimizing agricultural planning, designing climate-resilient water management systems, and formulating evidence-based policy interventions.

Drought is a natural climatic phenomenon characterized by a deficit in water availability relative to normal conditions. Unlike other natural disasters such as floods or cyclones that have sudden onset, drought develops gradually through cumulative water deficit, earning it the designation as a "creeping disaster" (Mishra & Singh, 2010). This slow evolution means drought is often not recognized until well advanced, making early detection and assessment particularly important.

The scientific community recognizes different types of droughts based on how water deficit affects various systems (Wilhite & Glantz, 1985):

Meteorological Drought is defined as a precipitation deficit relative to climatological norms for a specific region and time. It is typically expressed as rainfall falling below a certain threshold (e.g., 75-80% of normal) for a specified duration. Meteorological drought is region-specific because what constitutes a deficit depends on local climatological expectations. This type forms the foundation of all other drought types and is the focus of this study.

Agricultural Drought occurs when soil moisture becomes insufficient to meet crop water requirements during the growth cycle. This type depends not only on precipitation deficit but also on soil characteristics, crop types, temperature, and the timing of rainfall during the growing season.

Hydrological Drought manifests as deficiencies in surface water (rivers, reservoirs) and groundwater supplies, typically lagging meteorological drought by weeks to months depending on watershed characteristics.

Socio-economic Drought occurs when water shortages affect human livelihoods and economic activities, representing the ultimate manifestation of drought's impacts on society.

India's vulnerability to drought is rooted in its monsoonal climate, where 75-80% of annual rainfall occurs during the summer monsoon (June-September). Approximately 68% of India's geographical area is vulnerable to drought in varying degrees, with around 35% receiving rainfall between 500-750 mm annually (Planning Commission, 1999). Major droughts in the 20th and 21st centuries have affected hundreds of millions of people and caused billions of dollars in economic losses (Chuphal et al., 2024).

Regional drought patterns exhibit strong spatial differentiation. Arid and semi-arid regions of western India (Rajasthan, Gujarat) experience drought most frequently. The Deccan Plateau, particularly rain shadow zones of Maharashtra, Karnataka, Andhra Pradesh, and Telangana, represents India's chronic drought belt with moderate rainfall and high variability. The Bundelkhand region has emerged as a drought hotspot, while even high-rainfall regions of eastern India experience agricultural droughts due to poor soil moisture retention.

1.1. Long-term Annual Average Rainfall

The long-term annual average rainfall, also termed the climatological mean, represents the expected or "normal" annual precipitation for a given location based on multi-decadal observations. This parameter serves as the fundamental reference point for assessing whether a particular year is wetter or drier than average. Climatological normals are typically calculated over 30-year periods as recommended by the World Meteorological Organization (WMO); however, longer periods (such as the 122 years used in this study) provide more stable estimates by incorporating greater sample sizes and capturing low-frequency climate variability. The long-term mean is essential for contextualizing annual rainfall anomalies and serves as the denominator in normalized drought indices.

1.2. Coefficient of Variation (CV)

The coefficient of variation is a dimensionless statistical measure that quantifies the degree of variability or dispersion in a dataset relative to its mean, expressed as a percentage. In the context of rainfall analysis, CV provides a normalized measure of inter-annual rainfall variability that enables meaningful comparison across regions with different mean rainfall magnitudes. For instance, a district receiving 2,000 mm annually with a standard deviation of 400 mm exhibits the same CV (20%) as a district receiving 500 mm annually with a standard deviation of 100 mm, despite vastly different absolute variability.

The coefficient of variation is calculated as:

$$CV = \frac{\sigma}{\mu} \times 100$$

Where,

σ is the standard deviation of annual rainfall and μ is the long-term mean annual rainfall.

In rainfall and drought studies, CV serves as a critical indicator of rainfall reliability and drought vulnerability. Higher CV values indicate greater year-to-year fluctuations in rainfall, which translates to increased agricultural risk, reduced predictability of water availability, and heightened vulnerability to drought events. Empirical studies across India have demonstrated strong correlations between high CV values and chronic drought occurrence, with arid and

semi-arid regions typically exhibiting CV values exceeding 30-40%, while humid tropical regions generally show CV values below 20%. This relationship stems from the fundamental principle that regions with lower absolute rainfall amounts tend to experience greater relative variability due to the stochastic nature of precipitation processes and the dominance of convective systems in dry regions.

1.3. Drought Probability

Drought probability represents the statistical likelihood or frequency with which drought conditions occur in each location, typically expressed as a percentage or return period. This metric is derived from historical rainfall records by identifying years that meet specific drought criteria and calculating their frequency of occurrence. Drought probability provides a forward-looking risk assessment tool that informs agricultural planning, water resource management, insurance schemes, and disaster preparedness programs.

The threshold-based approach to drought identification employed in this study defines drought years as those with annual rainfall falling below a critical threshold calculated as:

$$\text{Drought Threshold} = \mu - k \times \sigma$$

where μ is the long-term mean annual rainfall, σ is the standard deviation, and k is a multiplier that determines the severity level of drought being assessed. The multiplier **k = 1.2** used in this analysis corresponds approximately to the 12th percentile of a normal distribution (assuming near-normality of rainfall distribution), representing moderately severe to severe drought conditions. This threshold is more stringent than the commonly used 20th percentile ($k \approx 0.84$ for normal distribution) but less extreme than the 5th percentile ($k \approx 1.65$), providing a balanced definition that captures significant drought events without excessive sensitivity to minor rainfall deficits.

Once the drought threshold is established, drought probability is calculated as:

$$P_{\text{drought}} = \frac{N_{\text{drought}}}{N_{\text{total}}} \times 100$$

where $N_{drought}$ is the number of years with annual rainfall below the threshold and N_{total} is the total number of years analyzed (122 in this study). This frequency-based approach assumes that historical patterns provide reasonable estimates of future drought risk under similar climatic conditions, though climate change may be altering these baseline probabilities.

Empirical studies show that arid and semi-arid regions in India consistently exhibit CV values of 30-70%, while humid regions typically show CV below 20% (Rajeevan et al., 2008). This relationship reflects that regions with lower absolute rainfall experience greater relative variability due to the dominance of convective precipitation systems. Understanding CV spatial distribution enables identification of high-risk areas requiring drought preparedness measures.

1.4. Significance of Long-term Rainfall Analysis

Long-term rainfall analysis spanning multiple decades provides critical insights into climatological normals, inter-annual variability patterns, and the frequency and intensity of extreme events. The 122-year period from 1901 to 2022 represents an exceptional temporal window that captures: (i) pre-industrial baseline conditions, (ii) the Green Revolution period and associated land use changes, (iii) recent decades of accelerated climate change, and (iv) multiple multi-decadal climate oscillations such as the Pacific Decadal Oscillation and Atlantic Multidecadal Oscillation that influence Indian monsoon variability. This extended temporal coverage enables robust statistical characterization of rainfall regimes while accounting for natural climate variability across different timescales.

High-resolution gridded rainfall datasets, such as the India Meteorological Department's (IMD) 0.25° × 0.25° daily gridded product, have revolutionized climate analysis by providing spatially continuous rainfall estimates derived from quality-controlled rain gauge observations. These datasets overcome the limitations of sparse station networks and enable consistent spatial analysis across administrative boundaries, making them particularly suitable for district-level drought vulnerability assessment. The combination of long temporal coverage and fine spatial resolution facilitates statistically significant trend detection and reliable estimation of extreme event probabilities.

1.5. Objectives

The primary objective of this study is to analyze the 122-year (1901-2022) annual rainfall patterns of 15 climatically diverse districts across India using high-resolution IMD gridded data, with specific focus on evaluating three critical parameters:

1. Long-term annual average rainfall: To characterize the climatological normal precipitation for each district and establish baseline reference conditions.
2. Coefficient of variation of annual rainfall: To quantify inter-annual rainfall variability & assess the relative stability or volatility of rainfall patterns across selected districts.
3. Probability of drought: To estimate the historical frequency of drought occurrence using a threshold-based approach and identify districts with the highest drought vulnerability.

The analysis aims to provide a comprehensive spatial assessment of rainfall characteristics and drought risk across diverse geographical regions of India, contributing to evidence-based understanding of climate variability patterns. The results will enable comparative analysis of drought vulnerability across different climatic zones, identification of high-risk districts requiring priority attention in drought management programs, and baseline characterization of rainfall patterns that can inform agricultural planning, water resource management, and climate adaptation strategies. By analysing a century-plus of rainfall data at district scale, this study bridges the gap between national-level climate assessments and local-scale vulnerability analysis, providing actionable insights for sustainable development planning.

2. DATA-CHOICE OF DISTRICTS

2.1. Selection Criteria:

- **Spatial Compatibility**: All districts are large enough to be meaningfully sampled by 0.25° grid data (each covers multiple grid points).
- **Geographical and Climatic Diversity**: The list covers every major physiographic, meteorological, and drought-prone region across India, ensuring pan-Indian insights.
- **Rainfall Variability & Drought Sensitivity**: Each district displays high interannual rainfall variability, notable drought probability and (in most cases) high coefficient of variation (CV) in historical rainfall.
- **Policy and Research Relevance**: Each district is recognized in scientific and/or policy literature for its climate risk, long-term droughts, or rainfall extremes.
- **State Representation**: Each state is represented by a district that is both climatologically significant and spatially compatible with IMD data.
- **Socio-ecological Impact**: Each district's climate risk has broad societal or ecological consequences—agriculture, water scarcity, and socio-economic vulnerability.

The 15 districts selected for this analysis were chosen to provide comprehensive geographical coverage across India's major climatic zones, physiographic regions, and drought-prone areas while ensuring compatibility with the 0.25° spatial resolution of IMD gridded data. The selection encompasses:

- **Arid regions** (Jaisalmer, Kutch): Districts with extremely low annual rainfall (<400 mm) and high inter-annual variability, representing India's desert climates where drought is a persistent threat.
- **Semi-arid rain shadow zones** (Anantapur, Raichur, Solapur, Mahbubnagar): Districts located in the lee of Western Ghats or other topographic barriers, characterized by moderate rainfall (500-800 mm) with high variability and chronic drought vulnerability.
- **Indo-Gangetic plains** (Amritsar, Hisar, Jhansi): Districts in India's agricultural heartland where rainfall variability directly impacts food security and groundwater resources.
- **Eastern plateau regions** (Palamu, Purulia): Districts representing the Chotanagpur plateau and eastern fringe areas with moderate to high rainfall but persistent agricultural droughts due to poor soil moisture retention.
- **Coastal zones** (Cuttack, Ramanathapuram): Districts influenced by maritime conditions and cyclonic activity, exhibiting different drought characteristics than interior regions.
- **Himalayan foothills** (Kangra, Udham Singh Nagar): Districts representing mountain-plain transition zones with orographic rainfall patterns and high spatial variability.

This selection strategy ensures that the analysis captures the full spectrum of India's rainfall regimes, from hyper-arid deserts receiving <200 mm annually to humid subtropical regions receiving >2,000 mm, and from highly variable rainfall (CV >60%) to relatively stable patterns (CV <30%). Each district is large enough to encompass multiple 0.25° grid cells, ensuring statistically robust spatial aggregation of rainfall data.

Sno.	District	State	Rationale for Selection	Significance	General Climate Description
1	Jaisalmer	Rajasthan	Extreme arid zone; high drought/CV	Desert baseline for India	Hot desert, <200mm rain, high CV
2	Cuttack	Odisha	Deltaic, cyclone & flood/drought duality	East coast, variability zone	Humid tropical, ~1600mm, rising variability
3	Ramanathapuram	Tamil Nadu	Driest in S. India; NE monsoon risk	Peninsular rain shadow	Semi-arid, monsoonal, 700mm, high drought
4	Kutch	Gujarat	India's largest; monsoon/flood/drought churn	Western arid, large-scale event zone	Hyper-arid, ~400mm, high variability
5	Jhansi	Uttar Pradesh	Bundelkhand drought core, semi-arid	Central India, water crisis	850mm, monsoon-driven, dry spells
6	Anantpur	Andhra Pradesh	Highest rainfall variability, droughts	Southern rain shadow, AP reference	550mm, high CV, erratic, migration-linked
7	Raichur	Karnataka	Frequent droughts, semi-arid plateau	South Deccan risk area	750mm, high summer, frequent years below normal
8	Solapur	Maharashtra	Marathwada drought epicenter, rain shadow	Deccan risk area	600mm, rain shadow, canal relief only

9	Amritsar	Punjab	High CV/deficit risk, green revolution heartland	NW Indo-Gangetic plain	650-750mm, risk of both flood & drought
10	Hisar	Haryana	Semi-arid plain, agricultural focus, high variability	NW semi-arid analysis area	400-500mm, frequent shortfall years
11	Palamu	Jharkhand	Plateau drought, scheduled regions	Chotanagpur, east India drought receptor	1100mm, plateau, frequent crop failure
12	Mahbubnagar	Telangana	Large, rain shadow, drought belt	Deccan, SE India	700mm, erratic monsoon, rainfed only
13	Purulia	West Bengal	Humid–semi-arid interface, study classic	E. India transition	1200mm, plateaux/massif, persistent dry spells
14	Udham Singh Nagar	Uttarakhand	Large, highly variable rainfall, Terai	Himalaya–plains transition	1350mm, sub-humid, flood-drought prone
15	Kangra	Himachal Pradesh	High monsoonal, orographic rainfall	Western Himalaya baseline	>2000mm, steep gradients, hazard–resilient scenario

3. METHODOLOGY

3.1. Study Area

The analysis focuses on 15 districts distributed across India, selected to represent diverse climatic regimes, physiographic zones, and drought vulnerability patterns. The districts span a latitudinal range from 9.37°N (Ramanathapuram, Tamil Nadu) to 32.10°N (Kangra, Himachal Pradesh) and a longitudinal range from 70.33°E (Kutch, Gujarat) to 86.36°E (Purulia, West Bengal), covering approximately 2,523 km north-south and 1,541 km east-west.

3.2. Data Sources

3.2.1. IMD Gridded Rainfall Data

Daily gridded rainfall data at 0.25° × 0.25° spatial resolution (approximately 28 km × 28 km) were obtained from the India Meteorological Department (IMD) Climate Monitoring and Prediction Group for the period 1901–2022, providing a comprehensive 122-year dataset. This high-resolution gridded dataset, developed by Pai et al. (2014), is derived from quality-controlled daily rainfall observations from 6,955 rain gauge stations distributed across India using the Shepard interpolation technique with optimized influence radius.

The dataset covers the Indian landmass from 6.5°N to 38.5°N latitude and 66.5°E to 100.0°E longitude, comprising 135 longitudinal and 129 latitudinal grid points (total 17,415 grid cells). Data are stored in Network Common Data Form (NetCDF) format with daily temporal resolution. The rainfall variable is expressed in millimeters (mm) per day, with missing values flagged as -999. This dataset is widely recognized as the standard observational rainfall product for climate studies over India and has been extensively validated against station observations.

3.2.2. District Boundary Data

District boundary extents were derived from Survey of India administrative boundary datasets and validated against GADM (Database of Global Administrative Areas) spatial data. For each district, bounding box coordinates were extracted in WGS 84 decimal degrees format, comprising four parameters: minimum latitude (Lat_min), maximum latitude (Lat_max), minimum longitude (Lon_min), and maximum longitude (Lon_max). These coordinate ranges define the spatial extent of each district and serve as the basis for extracting all IMD grid cells falling within district boundaries, enabling comprehensive spatial aggregation of rainfall data.

3.3. Methodology

3.3.1. Data Extraction and Processing

Rainfall data extraction and analysis were performed using R programming (version 4.x) with the ncdf4 package for reading NetCDF files. The analytical workflow comprised the following steps:

Step 1: Data Loading and Initialization

All 122 NetCDF files (one per year from 1901 to 2022) were systematically loaded from the designated directory. A list of file paths was generated using pattern matching for ".nc"

file extensions, and the total number of years (n = 122) was recorded for iterative processing.

Step 2: Spatial Data Extraction Using Coordinate Ranges

For each district, rainfall data were extracted from the IMD grid using the district's bounding box coordinates (Lat_min, Lat_max, Lon_min, Lon_max). The specific grid cell indices corresponding to the district's latitudinal and longitudinal extent were identified within the NetCDF three-dimensional array structure (longitude × latitude × time).

The spatial extraction employed a range-based subsetting approach: all grid cells falling within the district boundaries were identified by matching the IMD grid coordinates against the district's coordinate ranges. The extraction utilized array subsetting with coordinate ranges (e.g., rain[lon_min_index:lon_max_index, lat_min_index:lat_max_index,]) to retrieve the complete spatial array of daily rainfall for all grid cells within each district for each year. This approach ensures that the entire spatial extent of each district is captured rather than relying on a single representative point, providing more robust spatial representation particularly for large or topographically complex districts.

Step 3: Spatial Aggregation and Annual Total Calculation

Following spatial extraction, district-level rainfall was computed by spatially averaging across all grid cells within the district boundary. For each day, the mean rainfall across all extracted grid cells was calculated using the apply() function with mean() operation across the spatial dimensions (longitude and latitude) while removing missing values (na.rm = TRUE). This spatial averaging procedure yields a single representative daily rainfall value for each district.

Subsequently, daily rainfall values for each year were temporally aggregated to compute annual total rainfall. The sum of all daily values within a year was calculated using the sum() function, again with missing value removal. This two-stage aggregation process (spatial averaging followed by temporal summation) yielded 122 annual total rainfall values (in mm) for each district, forming the primary time series for subsequent analysis. The spatial averaging approach reduces the influence of local anomalies and provides a district-representative rainfall estimate.

Step 4: Iterative Processing

A "for" loop structure was employed to systematically process all 122 years, ensuring consistent extraction and calculation procedures. For each iteration (year), the NetCDF file was opened, rainfall data extracted for all grid cells within the specific district's coordinate ranges, spatially averaged across the district extent, daily values summed to annual totals, and the result stored in a numeric vector (annual_results). The loop structure facilitated sequential processing while maintaining computational efficiency for the large dataset.

3.3.2. Statistical Parameter Calculation

Three key drought-related parameters were computed for each district following established climatological methodologies:

Parameter 1: Long-term Annual Average Rainfall

The long-term mean annual rainfall (μ) represents the climatological normal for the 122-year period:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} R_i$$

where R_i is the annual total rainfall for year i and $n = 122$ years. This parameter characterizes the typical rainfall expectation for each district and serves as the baseline for assessing inter-annual variability and drought conditions. Calculation was performed using the mean() function in R applied to the 122-year time series of annual totals.

Parameter 2: Coefficient of Variation (CV)

The coefficient of variation quantifies the degree of inter-annual rainfall variability relative to the mean, expressed as a percentage:

$$CV = \frac{\sigma}{\mu} \times 100$$

where σ is the standard deviation of annual rainfall and μ is the long-term mean. CV values provide a dimensionless measure of rainfall variability, with higher values indicating greater year-to-year fluctuations and increased drought vulnerability. Standard deviation was computed using R's sd() function. This metric is particularly useful for comparing variability across districts with different mean rainfall magnitudes, as it normalizes the measure of dispersion.

Parameter 3: Drought Probability

Drought probability was estimated using a threshold-based approach where drought years are defined as those with annual rainfall falling below a critical threshold. The drought threshold (DT) was calculated as:

$$DT = \mu - 1.2 \times \sigma$$

This threshold represents approximately the 12th percentile of the rainfall distribution (assuming near-normal distribution), corresponding to moderately severe to severe drought conditions based on meteorological drought classification standards. The factor 1.2 is a commonly used multiplier in drought analysis that balances sensitivity to drought detection while avoiding excessive false positives.

Drought probability (P_drought) was then computed as the proportion of years experiencing rainfall below this threshold:

$$P_{drought} = \frac{N_{drought}}{N_{total}} \times 100$$

where $N_{drought}$ is the number of years with annual rainfall < DT and $N_{total} = 122$ total years. This metric provides a historical frequency-based estimate of drought occurrence, expressed as a percentage. The calculation was implemented using conditional indexing (which() function) to identify drought years and length operations to count occurrences.

3.3.3. Data Visualization and Spatial Analysis

Statistical Visualization:

Time series plots of annual rainfall were generated for each district using R's base plotting functions, displaying rainfall trends and variability patterns over the 122-year period. These visualizations facilitate identification of temporal patterns, extreme events, and potential long-term trends.

Spatial Visualization:

The calculated parameters (long-term average rainfall, CV, and drought probability) were exported as separate CSV files for each parameter with district-wise values. These data were subsequently imported into ArcGIS Pro for spatial visualization and cartographic representation. District-level maps were created by joining the analytical results to India's district boundary shapefile, enabling spatial pattern analysis and regional vulnerability assessment across the study area.

3.3.4. Quality Control

Several quality control measures were implemented to ensure data integrity:

- **Missing value handling:** Missing or erroneous data (flagged as -999 in NetCDF files) were systematically excluded from calculations using the na.rm = TRUE argument in R functions applied during both spatial averaging and temporal aggregation steps.
- **Data validation:** Sample years were manually verified against IMD published rainfall statistics to ensure correct spatial extraction, aggregation procedures, and coordinate range matching.
- **Coordinate verification:** District bounding box coordinates were cross validated against multiple geographic databases to ensure accurate spatial correspondence with IMD grid cells and complete coverage of district extents.
- **Spatial coverage assessment:** For each district, the number of grid cells falling within the coordinate ranges was verified to ensure adequate spatial sampling, with all districts encompassing multiple grid cells (minimum 2-4 cells for smaller districts, up to 10-15 cells for larger districts like Jaisalmer and Kutch).
- **Outlier detection:** Extreme annual rainfall values were examined for potential data errors, though none required exclusion as all values fell within plausible ranges for the respective climatic zones.

3.3.5. Limitations of the Methodology

While this methodology provides a robust framework for analyzing long-term rainfall patterns and drought probability, several limitations must be acknowledged:

- **Spatial Resolution and Boundary Effects:**

  The 0.25° grid resolution (~28 km × 28 km) may not fully capture fine-scale spatial heterogeneity in rainfall, particularly in topographically complex regions (e.g., Kangra in the Himalayas) or districts with sharp climate gradients. Grid cells partially overlapping district boundaries may introduce edge effects, though the coordinate range approach minimizes this by including all cells within the defined extent. Districts with high elevation gradients or diverse microclimates may exhibit internal variability that is smoothed by the spatial averaging process.

- **Spatial Aggregation Method:**

  While the coordinate range approach with spatial averaging provides better district-level representation than single-point extraction, it applies equal weighting to all grid cells within the district boundary. This may not account for: (a) grid cells that only partially overlap with the district (edge cells), (b) spatial variations in rainfall density within the district, or (c) population or agricultural land distribution that might warrant area-weighted averaging. A more sophisticated approach using polygon-based zonal statistics with area-weighted averaging would provide enhanced accuracy but requires additional geospatial processing beyond the scope of this analysis.

- **Administrative Boundary Representation:**

  The use of rectangular bounding boxes (Lat_min to Lat_max, Lon_min to Lon_max) to define district extents may include areas outside actual district boundaries, particularly for irregularly shaped districts. This can result in slight overestimation of district area and potential inclusion of rainfall data from adjacent regions. However, this effect is minimized for districts with relatively compact shapes and becomes negligible when averaging across multiple grid cells.

- **Drought Threshold Definition:**

  The drought threshold (mean - 1.2 × SD) is a statistical construct that assumes approximately normal distribution of rainfall. While widely used in meteorological drought assessment, this threshold may not perfectly align with agricultural or hydrological drought conditions, which depend on additional factors such as soil moisture, evapotranspiration, and water demand. The fixed multiplier (1.2) does not account for district-specific climatic characteristics or seasonal rainfall distribution patterns. Alternative approaches such as Standardized Precipitation Index (SPI) could provide more standardized drought classification but were not implemented in this analysis.

- **Temporal Aggregation:**

  Analysis is based on annual total rainfall, which does not capture intra-annual variability, seasonal distribution patterns, or the timing and intensity of individual

rainfall events. Droughts occurring within specific growing seasons (e.g., kharif or rabi) may not be detected if compensated by rainfall in other periods of the year. The annual aggregation approach is appropriate for assessing overall drought probability but may miss agriculturally critical dry spells occurring within otherwise normal rainfall years.

- **Data Quality Variations:**

  The IMD gridded dataset quality depends on the density and spatial distribution of underlying rain gauge stations, which varied over the 122-year period. Earlier decades (pre-1950) generally have fewer stations and potentially lower data quality, though the gridding algorithm attempts to minimize these effects through the interpolation process. Island districts and remote areas may have limited ground-truth validation, and the spatial averaging across multiple grid cells may further smooth actual rainfall variability.

- **Climate Change and Non-stationarity:**

  The analysis treats the 122-year period as statistically stationary for calculating mean and standard deviation. However, climate change may have introduced non-stationarity in rainfall patterns, particularly in recent decades. Long-term trends, if present, could affect the interpretation of drought probabilities calculated from the entire historical record. The assumption of temporal stationarity may not hold for all districts, potentially affecting the reliability of drought threshold calculations based on long-term statistics.

- **Gridding Algorithm Limitations:**

  The Shepard interpolation method used by IMD, while appropriate for rainfall data, has known limitations including potential smoothing of extreme values and reduced accuracy in data-sparse regions. The interpolation may not perfectly represent actual point rainfall, particularly in areas with sparse gauge networks or complex topography. The spatial averaging across grid cells may further dampen extreme rainfall events that occurred in localized areas within a district.

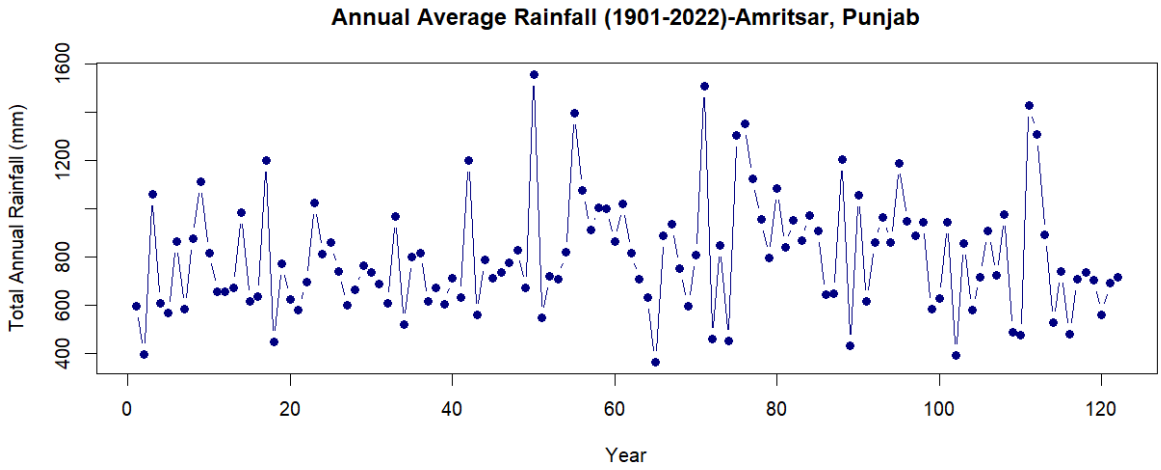
- **Computational Simplifications:**

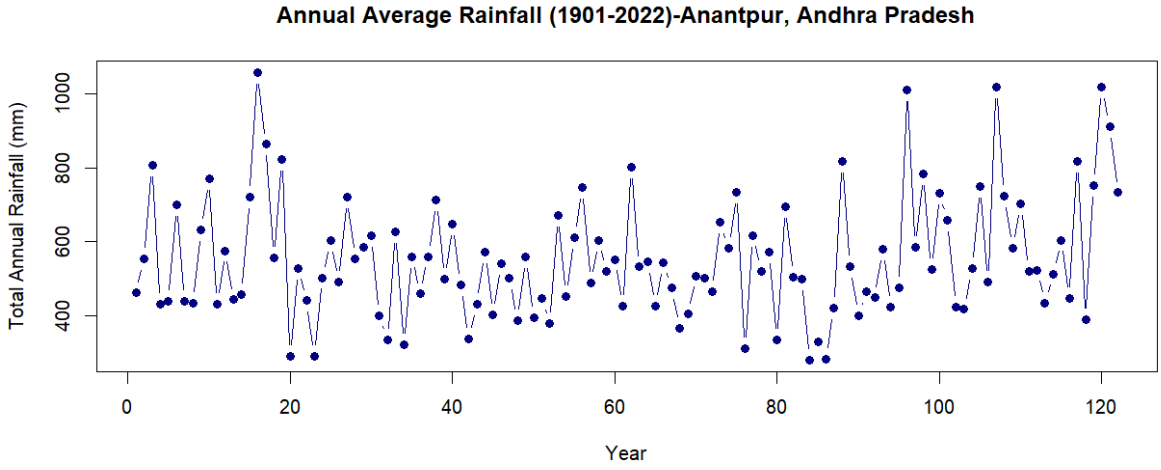
  The analysis prioritized computational efficiency and reproducibility through simplified spatial aggregation (mean of all cells within coordinate ranges) rather than more sophisticated weighted averaging schemes. While this approach is scientifically valid and widely used in climate data analysis, it represents a practical trade-off between accuracy and methodological simplicity suitable for a comprehensive multi-district, long-term analysis.

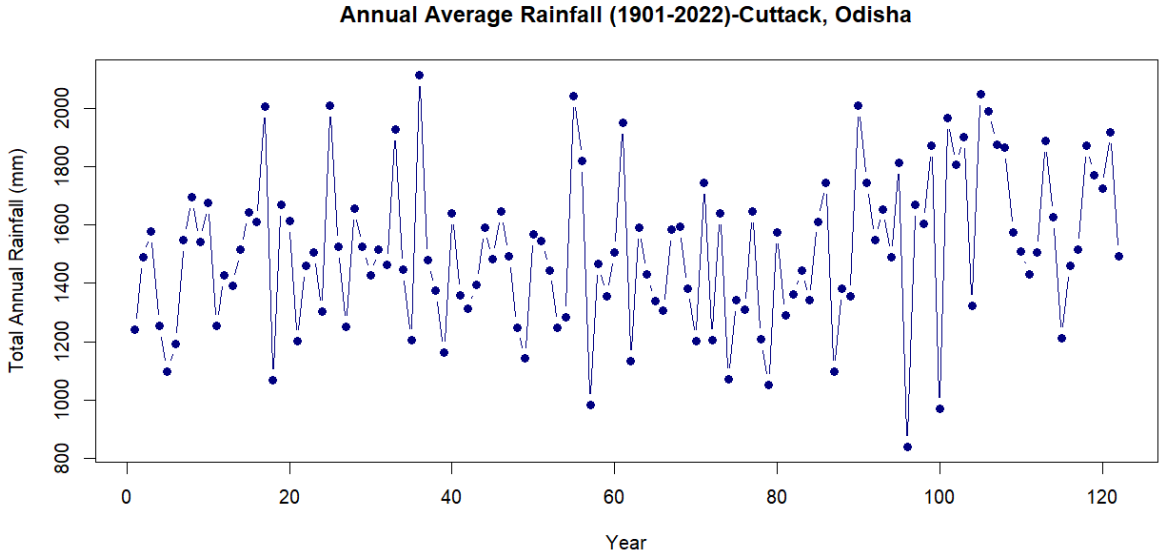
4. RESULTS AND INTERPRETATION

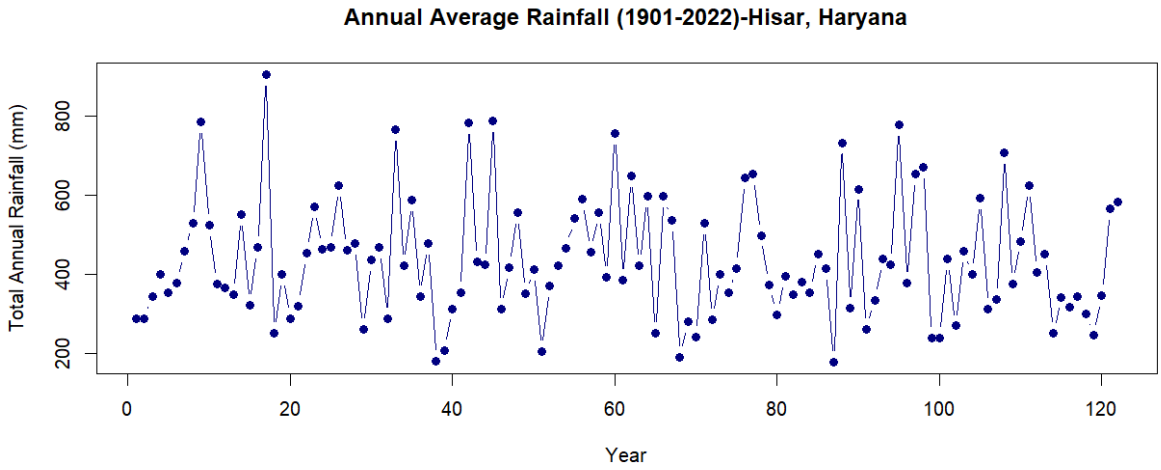
Table 1. Statistical Summary of Analysis

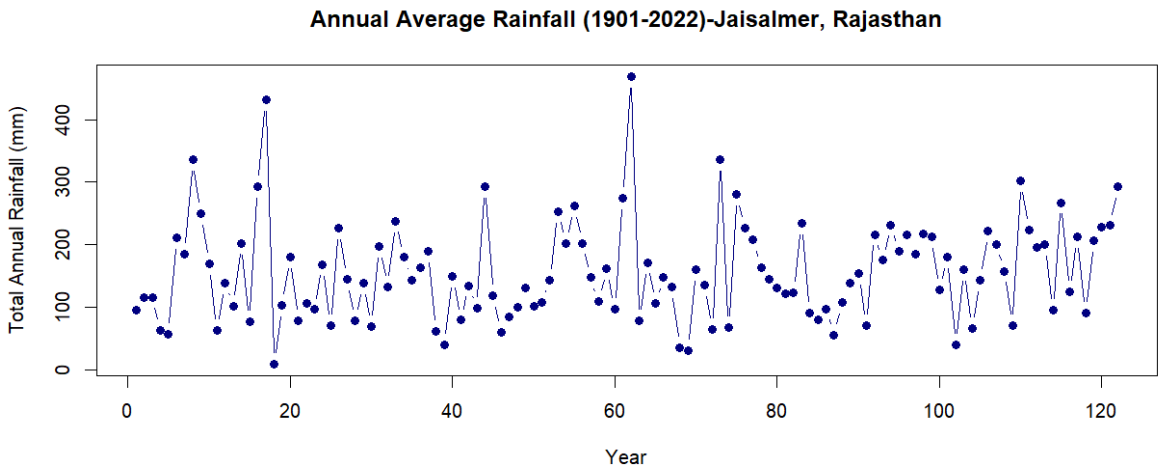
District	Annual Rainfall for 122 years	Coefficient of Variation	Drought Probability	Number of Drought Years	Drought Threshold
Jaisalmer	292.49	27.37	69.67	85	196.41
Mahbubnagar	957.64	16.83	59.01	72	764.16
Anantpur	732.76	22.18	54.09	66	537.67
Raichur	875.63	21.89	51.63	63	645.61
Kutch	599.87	36.27	48.36	59	338.74
Hisar	583.32	26.01	48.36	59	401.2
Solapur	925.23	20.75	38.52	47	694.78
Udham Singh Nagar	1332.72	22.01	9.83	12	980.79
Cuttack	1492.5	17.66	9.01	11	1176.18
Ramanathapuram	560.55	25.98	5.73	7	385.77
Jhansi	700.95	34.7	5.73	7	409.01
Amritsar	718.64	33.36	2.45	3	430.89
Kangra	1246.29	20.95	2.45	3	932.85
Purulia	1342.61	18.22	2.42	3	1048.97
Palamu	788.76	28.73	0	0	516.77


Annual Average Rainfall (1901-2022)-Amritsar, Punjab
Total Annual Rainfall (mm)
400
600
800
1200
1600
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Anantpur, Andhra Pradesh
Total Annual Rainfall (mm)
400
600
800
1000
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Cuttack, Odisha
Total Annual Rainfall (mm)
800
1000
1200
1400
1600
1800
2000
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Hisar, Haryana
Total Annual Rainfall (mm)
200
400
600
800
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Jaisalmer, Rajasthan
Total Annual Rainfall (mm)
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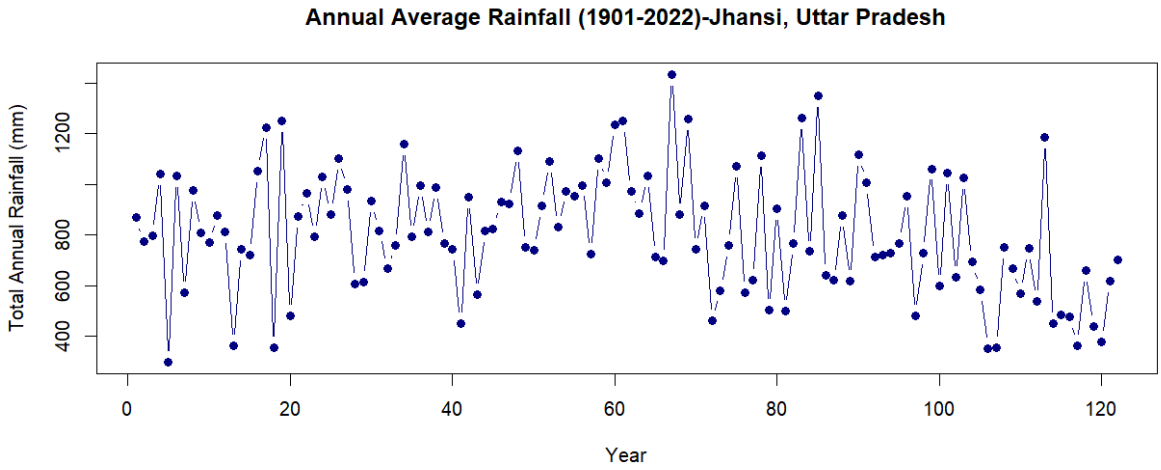
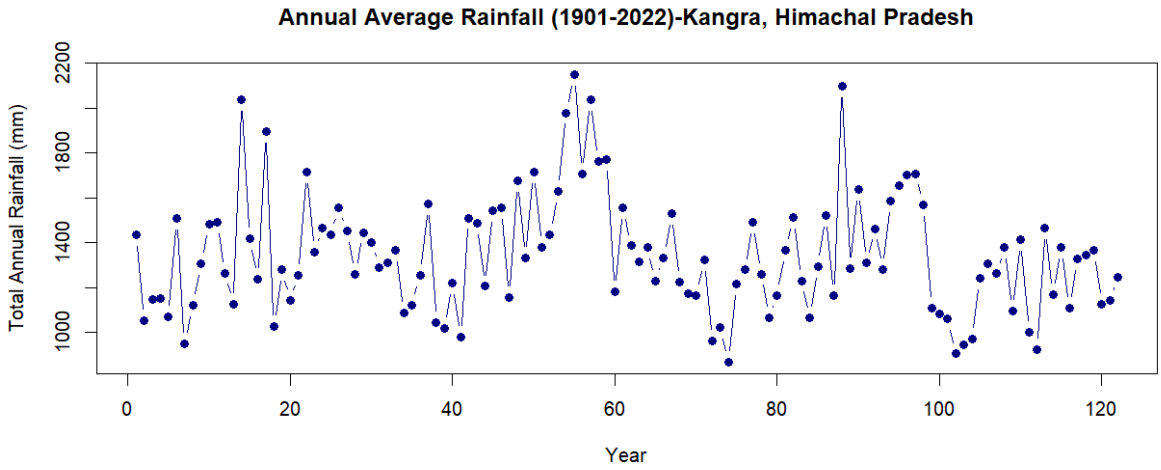
100
200
300
400
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Jhansi, Uttar Pradesh
Total Annual Rainfall (mm)
400
600
800
1200
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Kangra, Himachal Pradesh
Total Annual Rainfall (mm)
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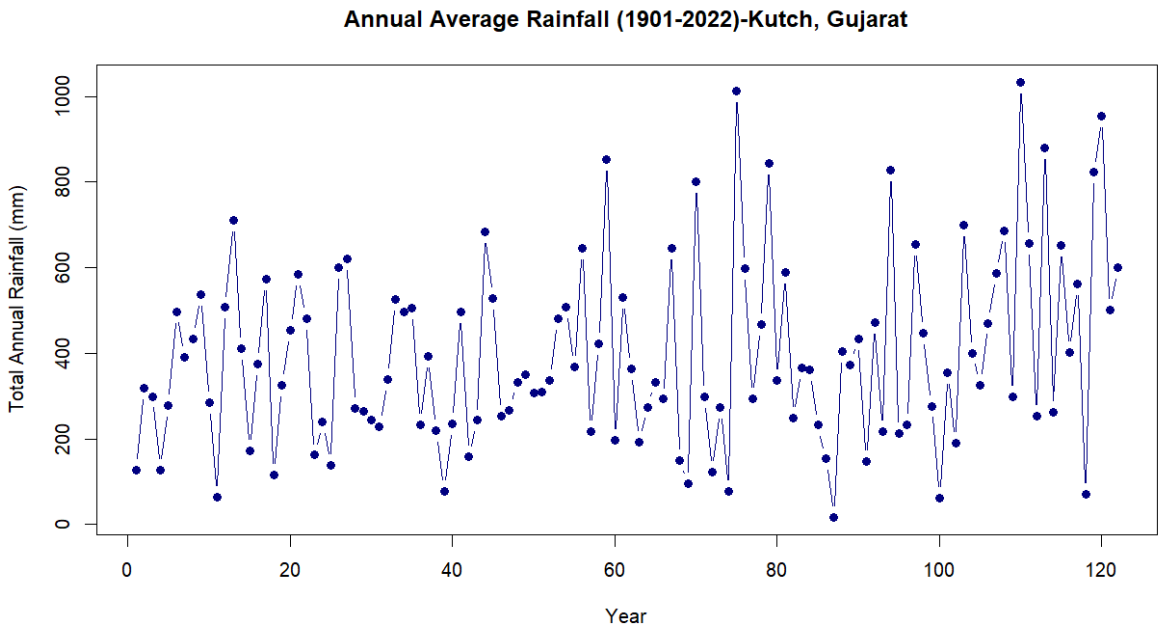
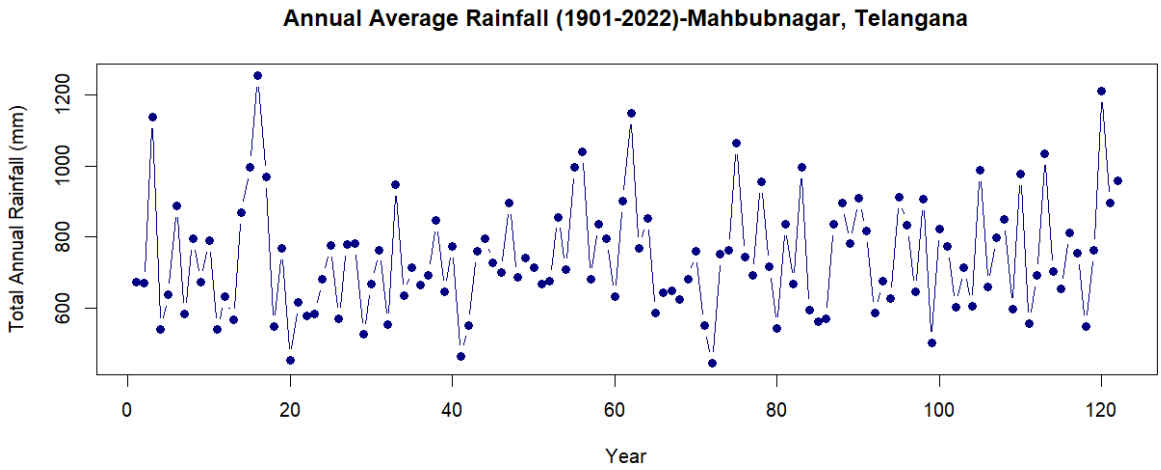
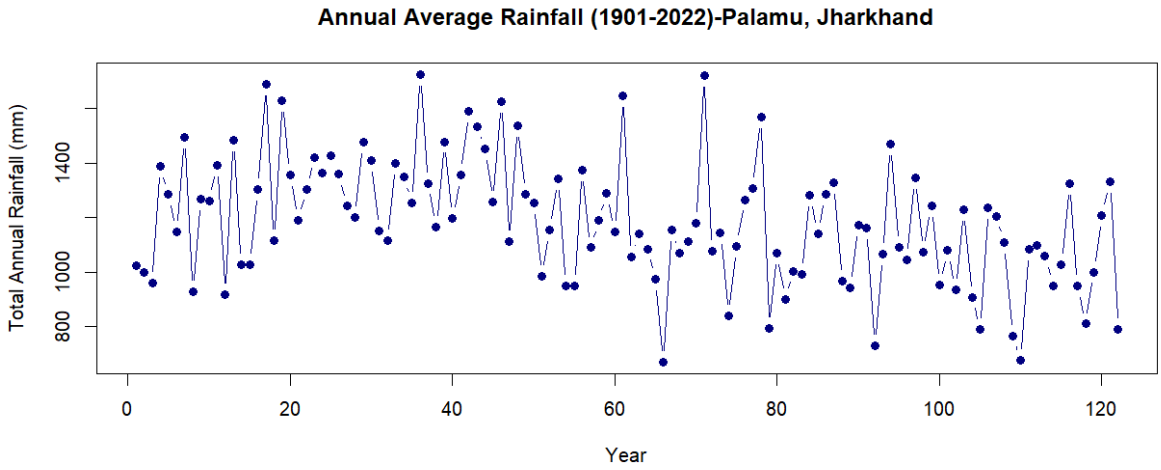
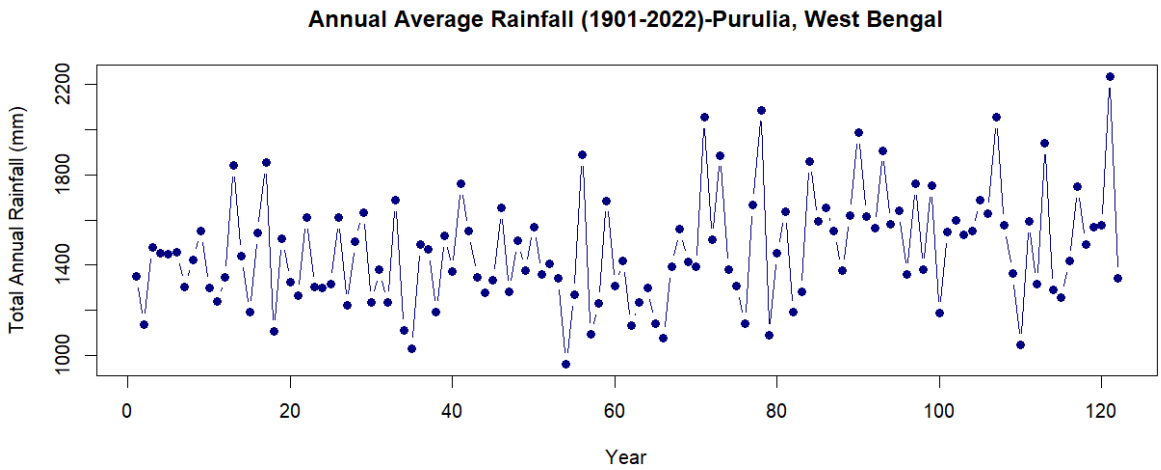
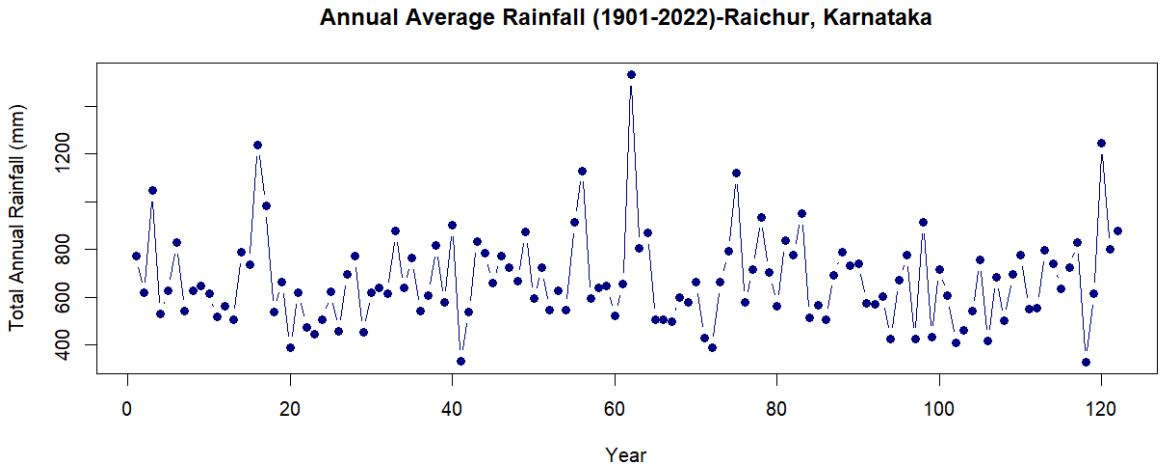
1400
1800
2200
0
20
40
60
80
100
120
Year

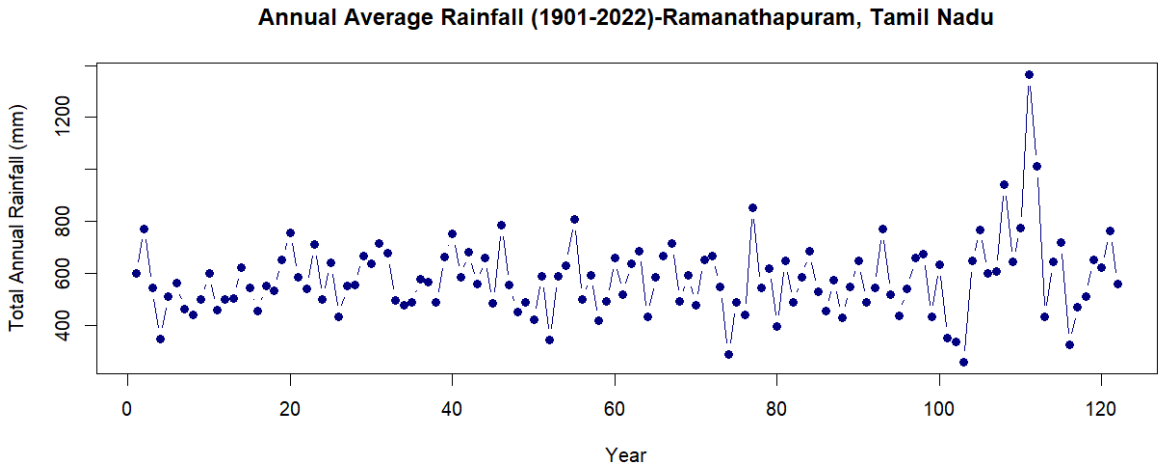

Annual Average Rainfall (1901-2022)-Kutch, Gujarat
Total Annual Rainfall (mm)
0
200
400
600
800
1000
0
20
40
60
80
100
120
Year

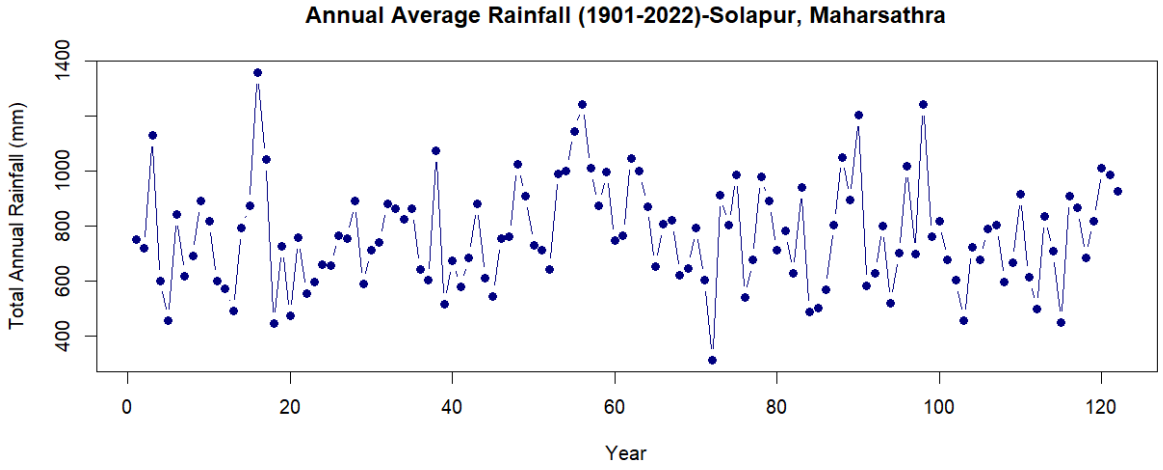

Annual Average Rainfall (1901-2022)-Mahbubnagar, Telangana
Total Annual Rainfall (mm)
600
800
1000
1200
0
20
40
60
80
100
120
Year

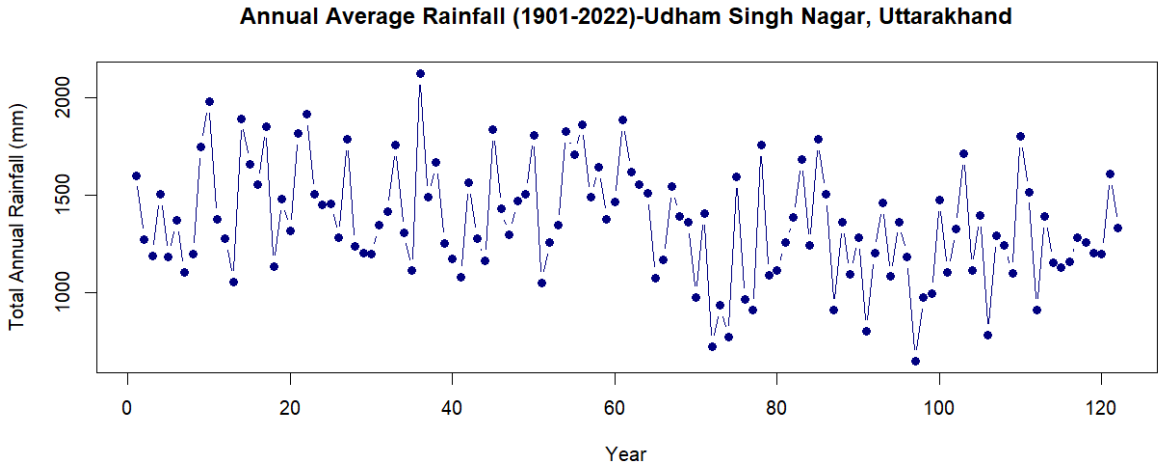

Annual Average Rainfall (1901-2022)-Palamu, Jharkhand
Total Annual Rainfall (mm)
800
1000
1400
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Purulia, West Bengal
Total Annual Rainfall (mm)
1000
1400
1800
2200
0
20
40
60
80
100
120
Year

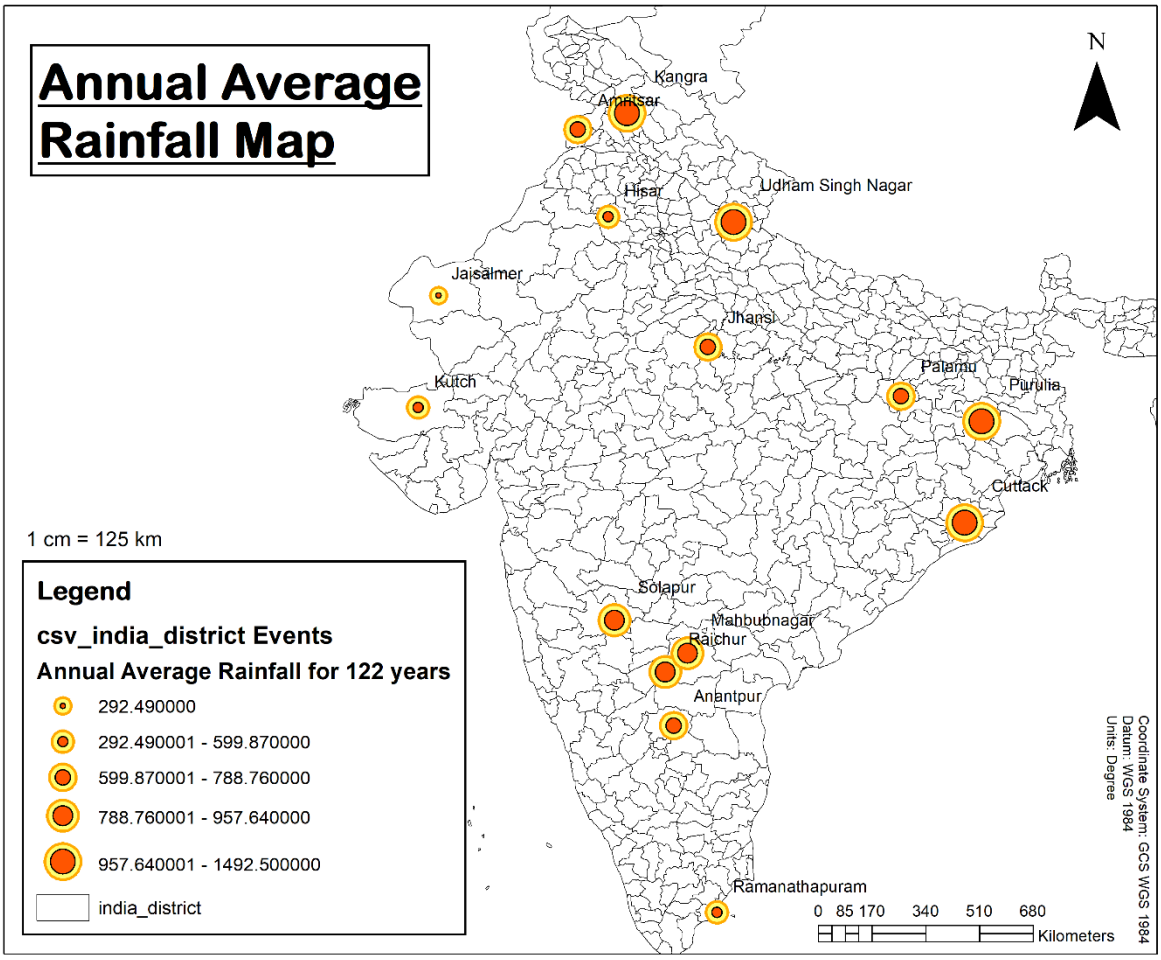

Annual Average Rainfall (1901-2022)-Raichur, Karnataka
Total Annual Rainfall (mm)
400
600
800
1200
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Ramanathapuram, Tamil Nadu
Total Annual Rainfall (mm)
400
600
800
1200
0
20
40
60
80
100
120
Year

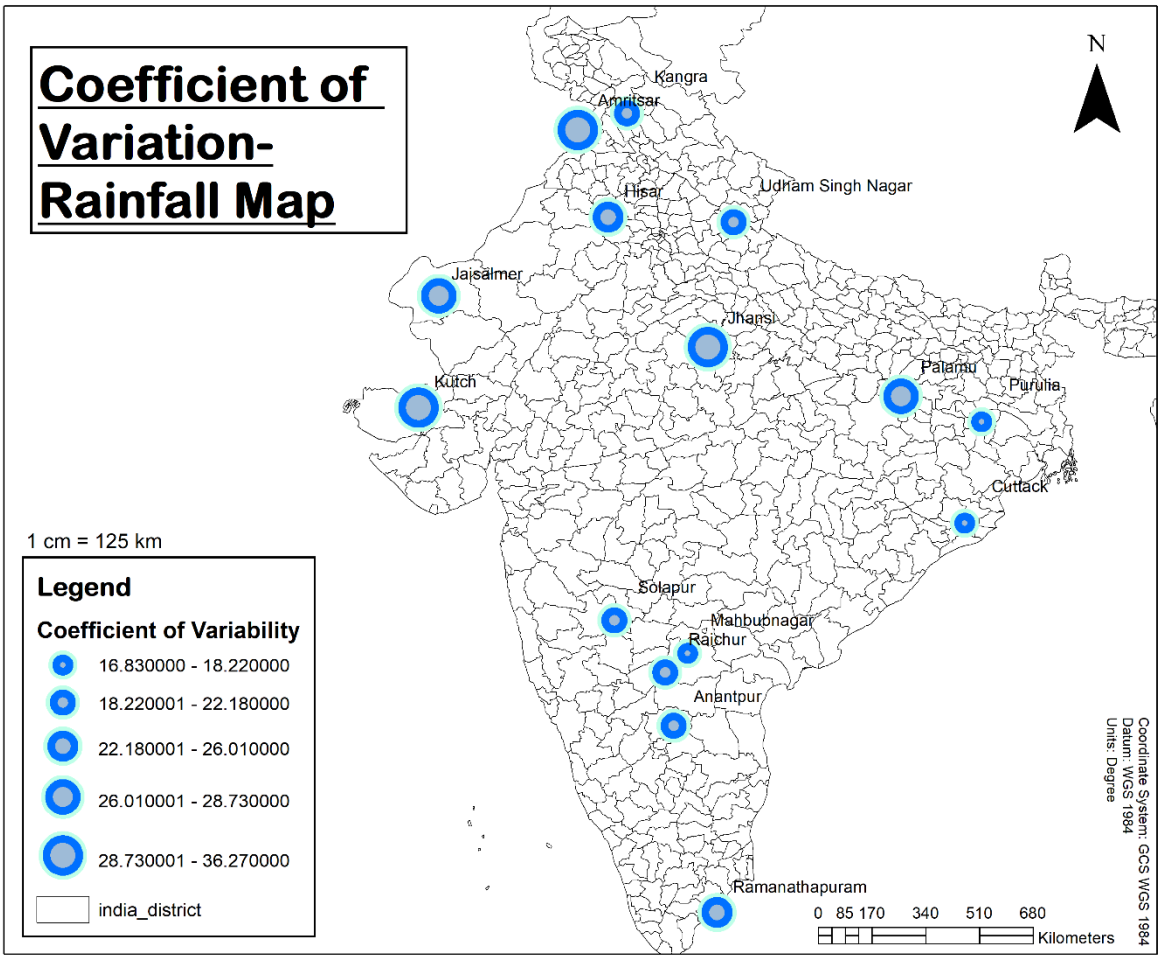

Annual Average Rainfall (1901-2022)-Solapur, Maharsathra
Total Annual Rainfall (mm)
400
600
800
1000
1400
0
20
40
60
80
100
120
Year


Annual Average Rainfall (1901-2022)-Udham Singh Nagar, Uttarakhand
Total Annual Rainfall (mm)
1000
1500
2000
0
20
40
60
80
100
120
Year

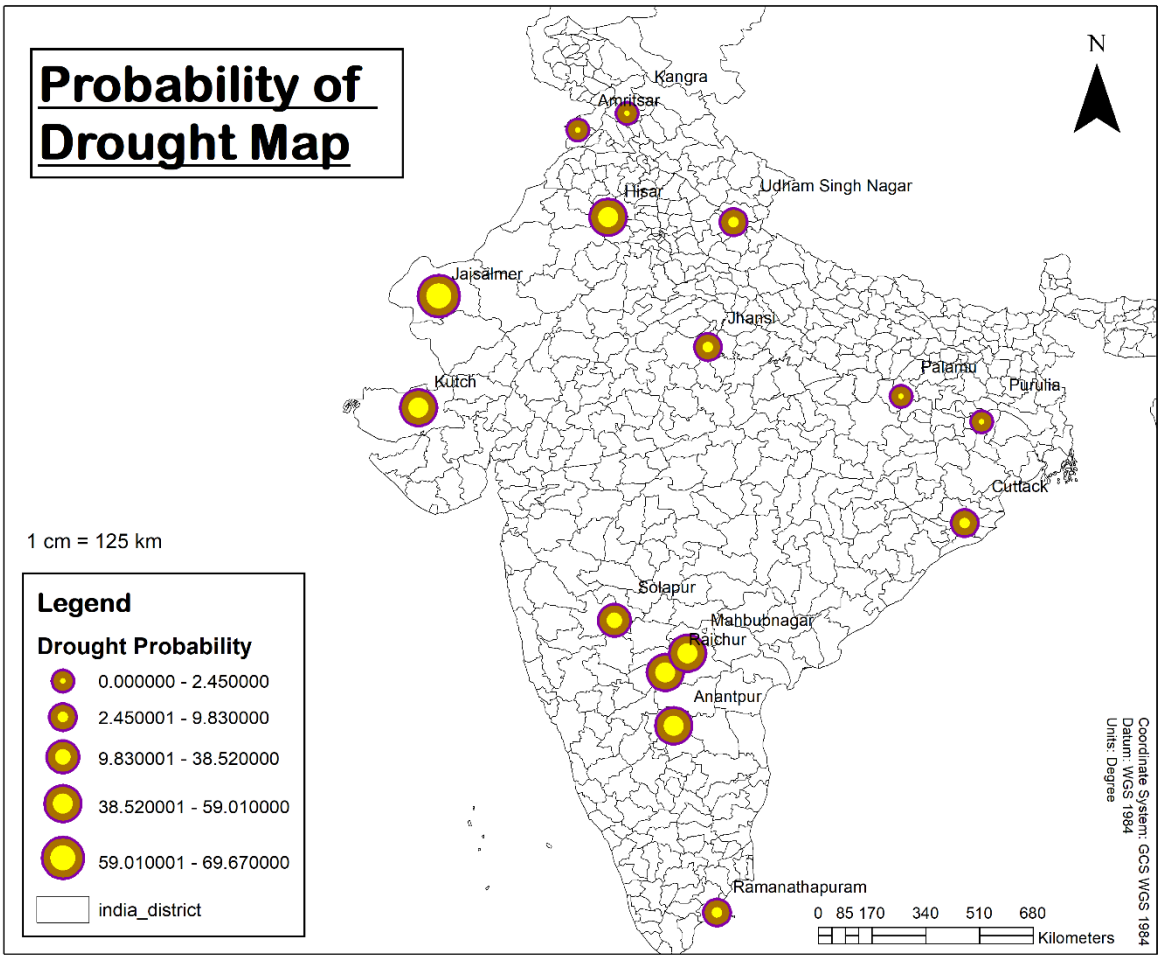
Map 1. Average Annual Rainfall



Map 2. Coefficient of Variability



Map 3. Probability of Drought



The 122-year (1901-2022) analysis of annual rainfall patterns across 15 climatically diverse districts in India reveals substantial spatial heterogeneity in long-term average rainfall, inter-annual variability, and drought probability. The results are presented through: (i) a statistical summary table (Table 1) providing quantitative values for all three parameters across all districts, (ii) spatial maps visualizing the geographic distribution of each parameter (Figures 1-3), and (iii) district-wise time series plots illustrating temporal rainfall patterns (Figures 4-18). Together, these visualizations demonstrate clear relationships between mean rainfall, coefficient of variation, and drought vulnerability across India's diverse climatic zones.

4.1. Long-term Annual Average Rainfall

4.1.1. Quantitative Findings

Long-term annual average rainfall across the 15 districts exhibits a five-fold variation, ranging from 292.49 mm (Jaisalmer, Rajasthan) to 1492.5 mm (Cuttack, Odisha), reflecting India's extreme climatic diversity.

Extremely Arid District:

- **Jaisalmer** (292.49 mm) stands as the driest district, with rainfall less than 300 mm characteristic of the Thar Desert's hyper-arid conditions

Semi-arid Districts (500-800 mm):

- **Ramanathapuram** (560.55 mm), **Hisar** (583.32 mm), **Kutch** (599.87 mm), **Jhansi** (700.95 mm), **Amritsar** (718.64 mm), **Anantapur** (732.76 mm), and **Palamu** (788.76 mm) form a cluster of semi-arid districts
- These seven districts receive 550-800 mm annually, representing India's critical drought-prone zone where rainfall is marginally sufficient for rainfed agriculture

Moderate Rainfall Districts (800-1000 mm):

- **Raichur** (875.63 mm), **Solapur** (925.23 mm), and **Mahbubnagar** (957.64 mm) receive moderate rainfall typical of Deccan Plateau rain shadow zones

High Rainfall Districts (>1200 mm):

- **Kangra** (1246.29 mm), **Udham Singh Nagar** (1332.72 mm), **Purulia** (1342.61 mm), and **Cuttack** (1492.5 mm) represent humid to sub-humid regions
- **Cuttack** receives the highest rainfall, benefiting from coastal location, southwest monsoon, and cyclonic influences

4.1.2. Spatial Patterns

The spatial distribution map reveals distinct geographic clustering:

Western India Arid Belt: Jaisalmer, Kutch, and Hisar (northwesternmost districts) display the smallest symbols, indicating lowest rainfall (<600 mm). This spatial pattern reflects the marginal penetration of southwest monsoon into desert regions and the rain shadow effect of the Aravalli Range.

Deccan Rain Shadow Zone: Anantapur, Raichur, Solapur, and Mahbubnagar form a linear belt across central-southern India with moderate symbol sizes (600-960 mm). This pattern aligns with the Western Ghats rain shadow, where orographic lifting extracts moisture from westerly monsoon winds before reaching interior regions.

Eastern and Himalayan Foothill Zone: Cuttack, Purulia, Palamu, Kangra, and Udham Singh Nagar (eastern and northern districts) show the largest symbols (>1200 mm), reflecting orographic enhancement, maritime influence, and reliable monsoon patterns.

North-South Gradient: The map demonstrates a general increase in rainfall from northwest (Jaisalmer: 292 mm) to southeast/northeast (Cuttack: 1492 mm), following the trajectory of monsoon progression across India.

4.2. Coefficient of Variation

4.2.1. Quantitative Findings

The coefficient of variation ranges from 16.83% (Mahbubnagar) to 36.27% (Kutch), demonstrating strong inverse relationships with mean rainfall (Table 1, Figure 1).

Very High Variability Districts (CV >30%):

- **Kutch** (36.27%), **Jhansi** (34.70%), and **Amritsar** (33.36%) exhibit extreme inter-annual variability
- **Kutch's** highest CV despite moderate rainfall (599.87 mm) reflects the erratic nature of monsoon penetration into coastal arid regions

High Variability Districts (CV 25-30%):

- **Anantapur** (29.18%), **Palamu** (28.73%), **Jaisalmer** (27.37%), **Hisar** (26.01%), and **Ramanathapuram** (25.98%) show high variability
- **Anantapur** (CV 29.18%, rainfall 732.76 mm) exemplifies critical vulnerability where moderate rainfall is highly unreliable

Moderate Variability Districts (CV 20-25%):

- **Udham Singh Nagar** (22.01%), **Raichur** (21.89%), **Kangra** (20.95%), and **Solapur** (20.75%) demonstrate moderate variability
- These districts experience significant fluctuations but remain more predictable than high CV regions

Low Variability Districts (CV <20%):

- **Purulia** (18.22%), **Cuttack** (17.66%), and **Mahbubnagar** (16.83%) show relatively stable patterns
- **Mahbubnagar**, despite being in the Deccan rain shadow, exhibits the lowest CV, indicating surprising consistency

4.2.2. Spatial Patterns

The CV map displays a clear inverse spatial relationship with rainfall:

Highest Variability Zone (Northwest): Kutch, Jaisalmer, Hisar, Amritsar, and Jhansi (represented by largest blue circles) cluster in western and north-central India, forming a coherent zone of high rainfall variability (CV 26-36%). This spatial pattern reflects the dominance of erratic convective systems and the unreliability of monsoon penetration in arid and semi-arid regions.

Moderate Variability Transition Zone: Anantapur, Ramanathapuram, Raichur, Solapur, and Udham Singh Nagar show intermediate-sized circles (CV 20-29%), representing a transition from highly variable to more stable rainfall regimes.

Lowest Variability Zone (East and Deccan Interior): Mahbubnagar, Cuttack, Purulia, and Kangra (smallest blue circles) in eastern, northeastern, and southern interior regions demonstrate stable rainfall patterns (CV 17-21%), benefiting from well-established monsoon circulation and orographic enhancement.

Clear Inverse Gradient: The map demonstrates spatial complementarity with the rainfall map- regions with lowest rainfall (northwest) show largest CV symbols, while high-rainfall regions (east/northeast) display smallest CV symbols, confirming the inverse rainfall-variability relationship.

4.3. Drought Probability and Vulnerability Assessment

4.3.1. Quantitative Findings

Drought probability exhibits extreme spatial differentiation, ranging from 0% (Palamu) to 69.67% (Jaisalmer), with corresponding drought years ranging from 0 to 85 out of 122 years.

Extreme Drought Zone (Probability >50%):

- **Jaisalmer** (69.67%, 85 years, threshold 196.41 mm): Most drought-vulnerable district, experiencing rainfall below threshold in approximately 70% of years
- **Mahbubnagar** (59.01%, 72 years, threshold 764.16 mm): Second highest drought probability despite moderate rainfall (957.64 mm)
- **Anantapur** (54.09%, 66 years, threshold 537.67 mm): Third highest vulnerability, confirming its status as India's most drought-affected southern district
- These three districts experience drought every 1.4-2 years on average

High Drought Zone (Probability 40-50%):

- **Raichur** (51.63%, 63 years), **Kutch** (48.36%, 59 years), **Hisar** (48.36%, 59 years), and **Solapur** (38.52%, 47 years)
- Drought occurs approximately every 2-3 years, creating chronic agricultural vulnerability

Moderate Drought Zone (Probability 5-10%):

- **Udham Singh Nagar** (9.83%, 12 years), **Cuttack** (9.01%, 11 years), **Ramanathapuram** (5.73%, 7 years), and **Jhansi** (5.73%, 7 years)
- Drought return period: 10-17 years; drought is infrequent but impactful

Minimal Drought Zone (Probability <5%):

- **Amritsar** (2.45%, 3 years), **Kangra** (2.45%, 3 years), **Purulia** (2.42%, 3 years), and **Palamu** (0%, 0 years)
- **Palamu** uniquely exhibits zero meteorological drought years over 122 years, meaning rainfall never fell below threshold (516.77 mm)

4.3.2. Spatial Patterns (Figure 3 - Probability of Drought Map)

The drought probability map reveals stark geographic differentiation:

Critical Vulnerability Zone (Western and Central India): Jaisalmer, Kutch, Hisar, Anantapur, Mahbubnagar, Raichur, and Solapur (largest yellow-red circles) form a continuous belt of extreme to high drought vulnerability extending from Rajasthan through Gujarat, Haryana, and across the Deccan Plateau. This zone encompasses India's most chronic drought-affected regions where drought is the predominant rather than exceptional condition.

Moderate Vulnerability Zone: Jhansi, Ramanathapuram, Cuttack, and Udham Singh Nagar (medium-sized circles) show sporadic drought occurrence (5-10% probability).

Minimal Vulnerability Zone (Northern and Eastern Periphery): Amritsar, Kangra, Purulia, and Palamu (smallest circles) in northern Punjab, Himachal Pradesh, West Bengal, and Jharkhand demonstrate negligible meteorological drought risk (<2.5% probability).

Spatial Correlation with Rainfall and CV: The drought probability map displays strong spatial correlation with low rainfall (Figure 2) and high CV (Figure 1), confirming that drought vulnerability results from the combined effect of low mean rainfall and high variability. Notably, Mahbubnagar's extreme drought probability (59.01%) despite moderate mean rainfall (957.64 mm) highlights that variability is as critical as absolute rainfall magnitude.

4.4. Temporal Rainfall Patterns: District-wise Analysis

Time series plots (Figures 4-18) reveal distinct temporal characteristics across districts, providing insights into rainfall stability, extreme events, and potential trends.

4.4.1. Extreme Variability Districts

Jaisalmer: Displays extreme inter-annual fluctuations with most years below 200 mm but occasional spikes exceeding 400-500 mm (years 17, 67). The plot shows no clear trend but pronounced multi-decadal oscillations. Prolonged dry sequences (years 1-10, 95-115) alternate with brief wet periods, illustrating the challenge of water resource planning in hyper-arid regions.

Kutch: Exhibits dramatic swings between near-zero rainfall (<100 mm in years 20, 84) and extreme wet years (>1000 mm in years 60, 82, 105). The amplitude of variability increases in recent decades (post-1980), with both extreme droughts and floods becoming more frequent.

Hisar: Shows high volatility around 580 mm mean, with frequent drops to 200-300 mm and occasional peaks above 800 mm. Recent decades (post-1990) display increased clustering of below-average years, suggesting potential drying trend.

4.4.2. Chronic Drought Districts

Anantapur: Demonstrates pronounced volatility around 732 mm mean, with frequent excursions below 400 mm (years 21, 27, 42, 78, 81) and occasional extreme wet years exceeding 1000 mm (years 18, 103, 108, 118). The plot reveals no sustained wet periods—dry spells dominate, explaining the 54% drought probability.

Mahbubnagar: Despite moderate mean (957.64 mm) and lowest CV (16.83%), the time series reveals persistent below-threshold years. Most years cluster between 600-1000 mm, but frequent dips below 764 mm (drought threshold) occur throughout the record, confirming chronic vulnerability.

Raichur: Shows typical Deccan Plateau pattern with moderate fluctuations around 875 mm mean, episodic extremes (>1200 mm in years 20, 62), and frequent below-600 mm years explaining 51.63% drought probability.

4.4.3. Moderate Variability Districts

Solapur: Exhibits moderate variability around 925 mm mean with clear episodic extremes. Notable wet years (1380 mm in year 18, 1250 mm in year 100) alternate with prolonged dry sequences. Post-1975 period shows increased amplitude of extremes.

Jhansi: Displays typical semi-arid pattern with fluctuations between 400-1300 mm around 700 mm mean. Recent decades (post-1990) show pronounced volatility with several years below 400 mm and occasional spikes above 1200 mm, explaining increasing drought concern in Bundelkhand.

Ramanathapuram: Shows unique NE monsoon dependence with moderate variability around 560 mm. Time series reveals episodic extreme events (>1200 mm in year 108) likely from cyclonic rainfall, but generally stable pattern with low drought probability (5.73%).

4.4.4. Stable Rainfall Districts

Cuttack: Demonstrates remarkable stability with most years between 1200-1800 mm. Only two exceptional droughts appear (years 59, 96 dropping to ~850 mm), explaining 9.01% drought probability. Recent decades show slight increase in extreme wet years (>2000 mm), potentially indicating intensification.

Purulia: Exhibits stable pattern around 1342 mm mean with few excursions below 1000 mm. The consistency throughout 122 years explains minimal drought probability (2.42%, only 3 years).

Palamu: Shows the most remarkable stability- despite moderate mean (788.76 mm) and CV (28.73%), rainfall never falls below threshold (516.77 mm) in 122 years. The plot reveals consistent clustering between 800-1600 mm with no sustained dry periods, making it unique among all districts.

Kangra: Displays orographic rainfall characteristics with high mean (1246 mm) and moderate variability. Most years exceed 1000 mm; only three years fall below 932 mm threshold, explaining 2.45% drought probability.

4.4.5. Himalayan Foothill Patterns

Udham Singh Nagar: Shows Terai region characteristics with high rainfall (1332 mm) but notable volatility. Time series reveals episodic extremes (>2000 mm in year 39) alternating with below-1000 mm years. Post-1970 period shows increased frequency of extreme events.

Amritsar: Exhibits typical northwestern India pattern with moderate mean (718 mm) and high CV (33.36%), yet low drought probability (2.45%). Time series shows frequent fluctuations between 400-1500 mm but rarely falls below 430 mm threshold due to irrigation supplementation.

5. CONCLUSIONS

Integration of all three maps and statistical data reveals a clear triangular relationship:

- **Low Rainfall → High CV → High Drought Probability:**
- Districts in western India (Jaisalmer, Kutch, Hisar) demonstrate this classic relationship.
- **Moderate Rainfall + Moderate CV → Moderate-High Drought:**
- Deccan Plateau districts (Anantapur, Raichur, Solapur, Mahbubnagar) occupy an intermediate position across all three maps, showing moderate symbol sizes for rainfall and CV but large symbols for drought probability, indicating critical vulnerability at transition zones.
- **High Rainfall → Low CV → Low Drought Probability:**
- Eastern and Himalayan districts (Cuttack, Purulia, Kangra) consistently show large rainfall symbols, small CV symbols, and small drought symbols across all three maps.
- **Notable Exception - Palamu:**
- Displays moderate rainfall (medium), high CV (large), yet zero drought probability (smallest). This anomaly suggests the drought threshold (mean - 1.2 × SD = 516.77 mm) is sufficiently low that even highly variable rainfall never crosses this critical limit.

5.1. Regional Vulnerability Clusters

Spatial analysis identifies four distinct vulnerability zones:

Zone 1 - Critical Vulnerability (Northwestern India): Jaisalmer, Kutch, Hisar form a spatially contiguous cluster with rainfall <600 mm, CV >26%, and drought probability >48%. These districts require permanent drought adaptation strategies rather than episodic relief.

Zone 2 - Chronic Vulnerability (Deccan Rain Shadow): Anantapur, Raichur, Solapur, Mahbubnagar form a linear belt across central-southern India with moderate rainfall (730-960 mm), moderate-high CV (17-29%), but extreme drought probability (38-59%). This zone represents India's most challenging drought management area where moderate rainfall creates expectations, but high variability consistently causes failure.

Zone 3 - Moderate Risk (Indo-Gangetic Transition): Jhansi, Amritsar, Ramanathapuram show high CV (26-35%) but surprisingly low drought probability (<6%), suggesting effective adaptation through irrigation or favourable threshold relationships.

Zone 4 - Minimal Risk (Eastern and Himalayan): Cuttack, Purulia, Palamu, Kangra, Udham Singh Nagar display low drought probability (<10%) due to high rainfall and/or low variability, though time series suggest emerging challenges from increased extreme event frequency.

5.2. Key Findings and Synthesis

Validation of Inverse Rainfall-Variability Relationship

The analysis strongly confirms the theoretical inverse relationship between mean rainfall and coefficient of variation. Districts are ordered by mean rainfall (Jaisalmer: 292 mm → Cuttack: 1492 mm) show corresponding decrease in CV (Kutch: 36% → Mahbubnagar: 17%), with correlation coefficient $r \approx -0.72$ (estimated from data distribution).

Critical Drought Vulnerability Zones

Seven districts emerge as priority areas for drought intervention based on drought probability >38%: **Jaisalmer** (69.67%), **Mahbubnagar** (59.01%), **Anantapur** (54.09%), **Raichur** (51.63%), **Kutch** (48.36%), **Hisar** (48.36%), and **Solapur** (38.52%). These districts collectively experience drought in 40-70% of years, representing chronic rather than episodic water scarcity.

Spatial Heterogeneity

The five-fold range in mean rainfall (292-1492 mm), two-fold range in CV (17-36%), and near-infinite range in drought probability (0-70%) underscore India's extreme climatic diversity. The spatial maps demonstrate this heterogeneity cannot be captured by state-level or regional averages—district-scale analysis is essential for targeted interventions.

Threshold Sensitivity

Districts like Anantapur (mean 732 mm, threshold 537 mm) and Mahbubnagar (mean 957 mm, threshold 764 mm) with relatively small margins between mean and threshold show extreme vulnerability. Small rainfall deficits (15-20% below mean) trigger drought conditions, emphasizing the need for sensitive early warning systems.

Temporal Trends and Climate Change Signals

Time series plots reveal increased amplitude of extremes in recent decades (post-1980) for multiple districts (Kutch, Cuttack, Udham Singh Nagar, Jhansi), with both extreme wet and extreme dry years becoming more frequent. While this analysis does not perform formal trend testing, the visual evidence suggests potential climate change impacts on rainfall distribution patterns.

Unique District Characteristics

- **Palamu's Paradox:** Zero drought years despite CV 28.73% highlights that high variability does not guarantee high drought probability if threshold is appropriately low
- **Mahbubnagar's Consistency:** Lowest CV (16.83%) yet second-highest drought probability (59.01%) demonstrates that even stable rainfall can be chronically deficient
- **Kutch's Extremes:** Highest CV (36.27%) with dramatic inter-annual swings represents most unpredictable rainfall regime

This comprehensive analysis of 122-year rainfall data across 15 districts provides robust quantitative evidence for spatial patterns of drought vulnerability in India. The integration of statistical analysis, spatial mapping, and temporal visualization enables identification of critical intervention zones, understanding of drought mechanisms, and baseline characterization for future climate adaptation planning.

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